

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459485.7292 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.212 +/- 0.063
Transit depth (Rp/Rs)^2	4.49 +/- 2.66 [%]
Orbital Inclination (inc)	84.1 +/- 2.77
Airmass coefficient 1 (a1)	1.02 +/- 0.57
Airmass coefficient 2 (a2)	0.17 +/- 0.65
Scatter in the residuals of the lightcurve fit is	52.43 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	6.5
Transit Duration (day)	0.064 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.212 ± 0.063 vs 0.1646 (deviation 0.58σ)
Duration fit vs published	0.064 d vs 0.0758 d (deviation 0.34σ)
Mid-time O–C	13.5 min
Depth SNR	2.6
Residual scatter	52.43 %

Light curve

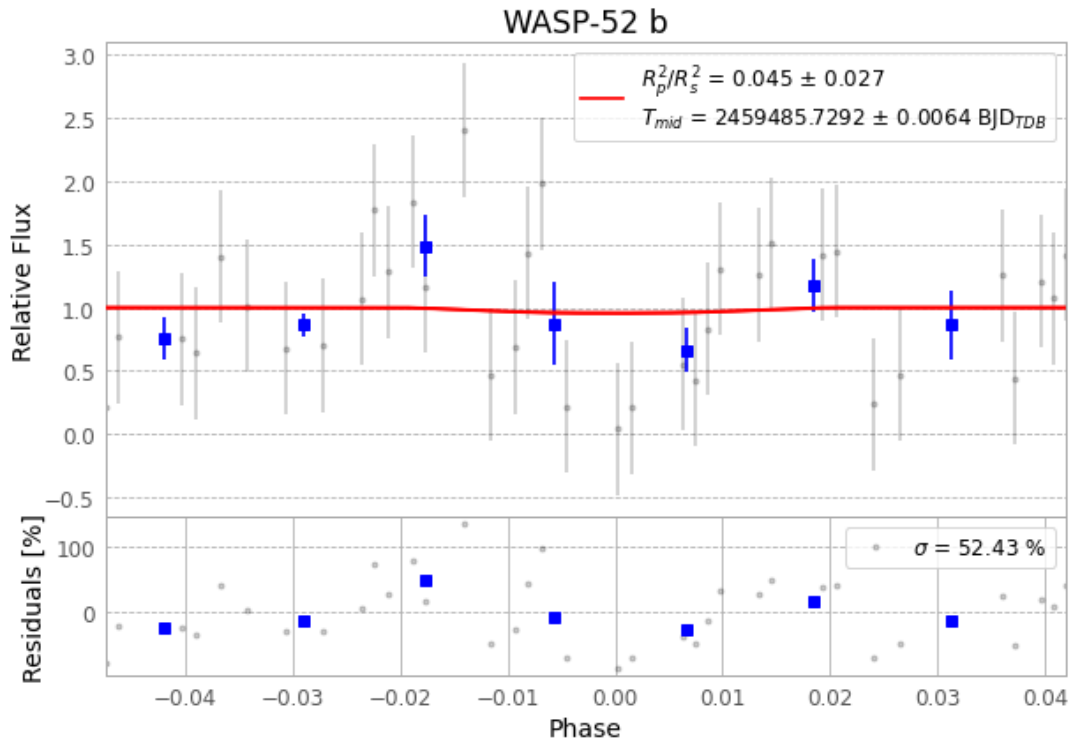


Figure 1 — FinalLightCurve_WASP-52 b_2021-09-27.png (SHA-256 11153cf00b4573a7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 138], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 687496a4443d01eb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-52210928031812.FITS → WASP-52210928070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-210928.log (67d52ede0106...), FinalLightCurve_WASP-52 b_2021-09-27.png (11153cf00b45...), FinalLightCurve_WASP-52 b_2021-09-27.csv (ed71d5e7f0ea...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 04:27Z.